

| 1.10 Vectors   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                |     |
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| 1.10a<br>1.10b | Vectors                            | <p>a) Be able to use vectors in two dimensions.<br/> <i>i.e. Learners should be able to use vectors expressed as <math>x\mathbf{i} + y\mathbf{j}</math> or as a column vector <math>\begin{pmatrix} x \\ y \end{pmatrix}</math>, to use vector notation appropriately either as <math>\overrightarrow{AB}</math> or <math>\mathbf{a}</math>.</i></p> <p><i>Learners should know the difference between a scalar and a vector, and should distinguish between them carefully when writing by hand.</i></p>                                                                                                                                                                                                                                                         | <p>b) Be able to use vectors in three dimensions.<br/> <i>i.e. Learners should be able to use vectors expressed as <math>x\mathbf{i} + y\mathbf{j} + z\mathbf{k}</math> or as a column vector <math>\begin{pmatrix} x \\ y \\ z \end{pmatrix}</math>.</i></p> <p><i>Includes extending 1.10c to 1.10g to include vectors in three dimensions, excluding the direction of a vector in three dimensions.</i></p> | MJ1 |
| 1.10c          | Magnitude and direction of vectors | <p>c) Be able to calculate the magnitude and direction of a vector and convert between component form and magnitude/direction form.</p> <p><i>Learners should know that the modulus of a vector is its magnitude and the direction of a vector is given by the angle the vector makes with a horizontal line parallel to the positive x-axis. The direction of a vector will be taken to be in the interval <math>[0^\circ, 360^\circ)</math>.</i></p> <p><i>Includes use of the notation <math> \mathbf{a} </math> for the magnitude of <math>\mathbf{a}</math> and <math> \overrightarrow{OA} </math> for the magnitude of <math>\overrightarrow{OA}</math>.</i></p> <p><i>Learners should be able to calculate the magnitude of a</i><br/> ( ... ) ( ... )</p> |                                                                                                                                                                                                                                                                                                                                                                                                                | MJ2 |

| OCR Ref.       | Subject Content               | Stage 1 learners should ...                                                                                                                                                                                                                                                                                                                                                 | Stage 2 learners additionally should ...                                                                                                                                                     | DfE Ref. |
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| 1.10d          | Basic operations on vectors   | <p>d) Be able to add vectors diagrammatically and perform the algebraic operations of vector addition and multiplication by scalars, and understand their geometrical interpretations.</p> <p><i>i.e. Either a scaling of a single vector or a displacement from one position to another by adding one or more vectors, often in the form of a triangle of vectors.</i></p> |                                                                                                                                                                                              | MJ3      |
| 1.10e          | Position vectors              | <p>e) Understand and be able to use position vectors.</p> <p><i>Learners should understand the meaning of displacement vector, component vector, resultant vector, parallel vector, equal vector and unit vector.</i></p>                                                                                                                                                   |                                                                                                                                                                                              | MJ4      |
| 1.10f          | Distance between points       | <p>f) Be able to calculate the distance between two points represented by position vectors.</p> <p><i>i.e. The distance between the points <math>a\mathbf{i} + b\mathbf{j}</math> and <math>c\mathbf{i} + d\mathbf{j}</math> is <math>\sqrt{(c-a)^2 + (d-b)^2}</math>.</i></p>                                                                                              |                                                                                                                                                                                              |          |
| 1.10g<br>1.10h | Problem solving using vectors | <p>g) Be able to use vectors to solve problems in pure mathematics and in context, including forces.</p>                                                                                                                                                                                                                                                                    | <p>h) Be able to use vectors to solve problems in kinematics.</p> <p><i>e.g. The equations of uniform acceleration may be used in vector form to find an unknown. See section 3.02e.</i></p> | MJ5      |